

PROPOSTA DI GRANT GIOVANI IN CSN5 - Anno 2026

Acronimo:	PRISM
Titolo completo:	Precision Room-temperature Instrumentation for high-energy x-ray absorption Spectroscopy Measurements
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Unità partecipanti:	INFN Sezione di Frascati

Abstract (2000 caratteri)

X-ray absorption spectroscopy (XAS) is a powerful element-selective technique for probing the electronic and atomic structure of materials. While synchrotron and free-electron laser facilities provide outstanding performance, their limited accessibility motivates advanced laboratory instrumentation. A major challenge is extending fluorescence-mode XAS beyond 20 keV, where silicon detectors lose efficiency and germanium detectors require cryogenic operation. This limits studies of elements such as Mo, Rh, Ag and Cd, whose K-edges are relevant to catalysis, electrochemistry, advanced alloys and energy materials. PRISM will develop a high-resolution laboratory hard X-ray spectrometer combining advanced crystal optics with a multichannel cadmium zinc telluride (CZT) detector for precision fluorescence spectroscopy up to 50 keV. Building on my expertise in crystal-based X-ray spectroscopy gained within the VOXES project and SIDDHARTA-2 experiment, PRISM will develop detector architectures, low-noise front-end electronics, calibration procedures and spectral reconstruction algorithms to achieve sub-keV energy resolution, high quantum efficiency (>90% up to ~40 keV) and excellent SNR performance at room temperature. The instrument will be validated through fluorescence XAS measurements on reference materials with K-edges above 20 keV, demonstrating accurate edge reconstruction and XANES spectra within laboratory-compatible acquisition times. The resulting capability will enable extended in situ and operando studies of slowly evolving catalytic, electrochemical and energy-related processes. Beyond this proof of concept, PRISM will establish a new detector platform for next-generation laboratory hard X-ray spectroscopy, strengthening INFN expertise in semiconductor radiation detectors, precision X-ray instrumentation and crystal spectroscopy. The technology will complement future photon-source infrastructures such as EuAPS and support research and industrial applications.

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1 State of the art (4000 caratteri)

A central challenge in materials characterization is determining how electronic and local atomic structure evolve under realistic conditions. Changes induced by irradiation, processing, phase transformations, chemical reactions or device operation may be transient, reversible or environment-dependent, and can therefore be missed by ex-situ measurements [1]. This requires in-situ and/or operando techniques capable of probing thick, heterogeneous or diluted samples while retaining sensitivity to selected elements and their local coordination [2,3]. Hence, there is a broad need for non-destructive, element-selective and bulk-sensitive probes compatible with complex environments and long acquisition campaigns.

X-rays are particularly suited for such investigations because of their high penetration depth with respect to other optical techniques, enabling access to the bulk of samples. By tuning the photon energy of a selected element, X-ray spectroscopies provide element- and orbital-specific information, even in chemically complex and heterogeneous systems. These properties make X-ray methods well-suited to non-destructive in situ and operando studies [4].

Among X-ray techniques, X-ray absorption spectroscopy (XAS) is particularly suited to this purpose because it measures the energy-dependent absorption coefficient across an element-specific absorption edge [5,6]. The near-edge region, the XANES, is sensitive to oxidation state, electronic structure and coordination geometry, whereas the extended region up to 1 keV above the edge, the EXAFS, provides quantitative information on the local atomic environment, including interatomic distances, coordination numbers and structural disorder. XAS can be measured in transmission, by recording the attenuation of the incident beam through the sample, or in fluorescence, by detecting the characteristic X-rays emitted after absorption. While transmission is generally preferred for homogeneous samples of suitable thickness, fluorescence detection is essential for diluted absorbers, thick specimens and complex sample environments, where the transmitted intensity is too weak or difficult to access.

Synchrotron facilities are the reference platforms for XAS, as their high brilliance, broad energy tunability and dedicated beamlines enable high-quality spectra with excellent signal-to-noise ratio over a wide energy range. X-ray free-electron lasers further provide ultrashort and intense X-ray pulses, opening access to femtosecond time-resolved experiments and transient states. However, access to these large-scale facilities is limited and competitive, and the allocated beamtime is often unsuitable for routine characterization, long experimental campaigns, or studies involving hazardous and radioactive samples [7].

These limitations have driven renewed interest in laboratory XAS systems based on conventional X-ray tubes [8]. Although laboratory sources provide substantially lower brilliance than synchrotrons, they offer continuous access, flexible sample handling and long measurement times, making them attractive for routine analysis, method development and slowly evolving in situ or operando processes. Recent advances in source brightness, crystal optics and detection technology have enabled an increasing number of XAS and XES measurements to be transferred back to the laboratory.

Within INFN, VOXES [9] has established expertise in high-resolution von Hamos spectroscopy for extended and dilute X-ray sources. The instrument currently operates mainly in the 5–20 keV range and has recently been upgraded with motorized positioning, energy-dispersive monitoring and mechanical compatibility with laboratory XAS measurements [10].

Despite this progress, laboratory fluorescence-XAS at energies above approximately 20 keV remains challenging. Silicon-based detectors progressively lose effectiveness in the hard-X-ray range, whereas high-purity germanium detectors offer excellent spectroscopic performance but require cryogenic cooling. Conversely, high-Z compound semiconductors such as CdTe and CdZnTe provide high

hard-X-ray absorption efficiency and room-temperature operation, although their use requires careful control of charge transport, detector segmentation, electronic noise and spectral reconstruction.

As a consequence, the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd remain difficult to investigate by laboratory fluorescence-XAS under realistic experimental conditions. There is therefore a clear need for a compact platform combining high-resolution crystal optics with efficient, room-temperature hard-X-ray detection. Addressing this technological gap would extend laboratory XAS beyond 20 keV and establish a complementary capability between conventional laboratory instruments, synchrotron facilities and future compact photon sources.

2 Goals and Expected Outcome (2000 caratteri)

The **main goal** of the project is **to develop and validate a room-temperature laboratory platform for fluorescence-mode X-ray absorption spectroscopy (f-XAS) above 20 keV**. Source, crystal optics, sample geometry, CZT detection, electronics and simulations will be optimized as a single measurement chain to obtain chemically meaningful XANES spectra from samples for which transmission is impractical.

The project is organized into the following specific **goals (G)** and **expected outcomes (EO)**:

G1 – Establish a validated design and predictive model of the complete instrument.

EO1 – An integrated simulation workflow and a frozen experimental design, validated at the Ag K-edge and at one energy of at least 40 keV. Predictions and measurements will agree within 20% for detected rate and within 10% for spot size and energy resolution.

G2 – Develop and characterize a multichannel CZT detector operating at room temperature.

EO2 – A calibrated system with intrinsic efficiency above 90% at 40 keV, energy resolution of 1 keV FWHM or better over 20–40 keV, dead time below 10% and residual pile-up below 5% at the selected operating rate.

G3 – Validate the complete platform on reference compounds with different chemical states.

EO3 – Quantitative Ag K-edge XANES with effective XAS resolution of 10 eV FWHM or better, edge-position reproducibility within 1 eV, edge-step SNR of at least 20 and acquisition time below the 12 h laboratory benchmark. The normalized spectral deviation from reference data will be below 5%.

G4 – Demonstrate f-XAS on at least one thick, diluted or operando-compatible sample.

EO4 – Discrimination of at least two chemical states through an edge shift of at least 1 eV and/or a normalized white-line variation of at least 5%, with a significance of at least 3σ and an acquisition time below 12 h per spectrum. The measured operating domain will be reported in terms of concentration, sample geometry and acquisition time.

3 Methodology (10000 caratteri)

XAS measures the energy dependence of the linear absorption coefficient, $\mu(E)$, across an element-specific core-level absorption edge (see Figure 1). When the incident photon energy reaches the binding energy of a core electron, the absorption probability rises sharply and the resulting photoelectron probes both the unoccupied electronic states and the surrounding atomic structure. The near-edge region (XANES) is sensitive to oxidation state, orbital occupancy and local symmetry,

whereas the oscillations above the edge (EXAFS) provide information on coordination numbers, interatomic distances and structural disorder. Laboratory XAS becomes increasingly demanding at hard-X-ray energies because the usable flux from conventional sources decreases and EXAFS requires accurate measurements over a broad energy interval. PRISM will use the Ag K-edge at 25.5 keV as its benchmark, first establishing a robust XANES protocol and then quantifying the EXAFS range achievable within laboratory-compatible acquisition times.

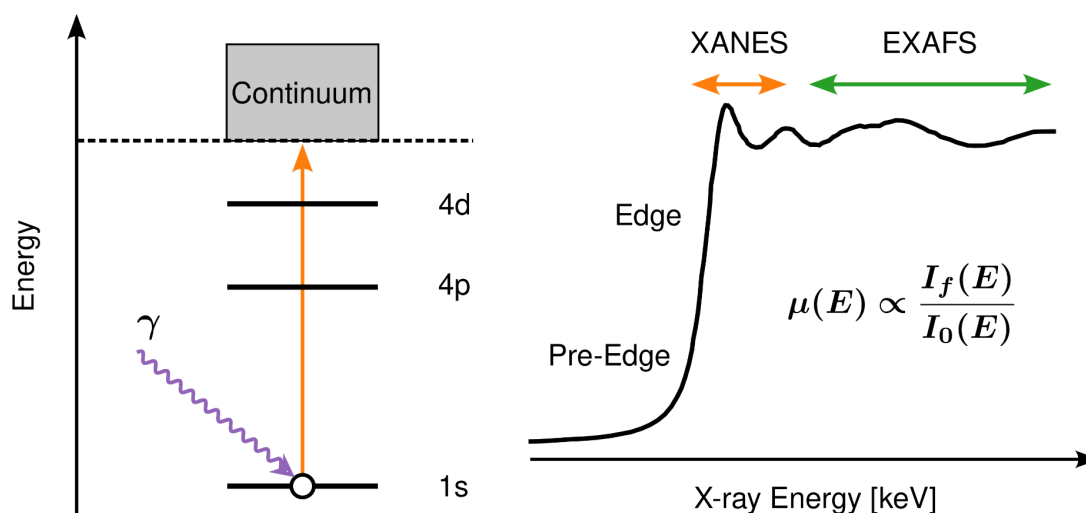


FIGURE 1. Principles of XAS illustrated for the Ag K-edge. Left: schematic of the electronic transitions involved in X-ray absorption. When the incident photon energy exceeds the Ag K-edge (≈ 25.5 keV), a 1s core electron is excited into the continuum, creating a core hole. Right: qualitative evolution of the absorption coefficient, $\mu(E)$, as a function of photon energy, highlighting the XANES (X-ray Absorption Near Edge Structure) and EXAFS (Extended X-ray Absorption Fine Structure) regions. XANES is primarily sensitive to the oxidation state and local electronic structure, while EXAFS provides quantitative information on the local atomic environment, including coordination numbers and interatomic distances.

The absorption coefficient $\mu(E)$ can be acquired in transmission or fluorescence mode. In a dispersive transmission arrangement, such as that being explored with VOXES, the broad source spectrum passes through the sample and a monochromator crystal spatially separates the transmitted energies onto a position-sensitive detector. Comparison of the spectra acquired with (I) and without (I_0) the sample provides $\mu(E)$. This approach is efficient but requires a homogeneous sample with an appropriate transmission thickness. PRISM will instead operate primarily in fluorescence mode. A crystal placed before the sample will select and focus one incident energy at a time, while an energy-resolving detector positioned approximately perpendicular to the incident beam will measure the characteristic fluorescence emitted following core-hole creation. For the Ag K-edge (see Figure 1), $I_f(E)$ will be extracted from the Ag $K\alpha$ line complex at approximately 22.2 keV. This geometry extends XAS to thick, dilute, heterogeneous or geometrically constrained samples that are impractical in transmission. In the thin or dilute limit, after background, live-time and detector-response corrections, the absorption spectrum is proportional to the ratio $\mu(E) \propto I_f(E) / I_0(E)$ where $I_0(E)$ is the incident monochromatic intensity.

The incident energy, impinging on the sample, will be selected from the X-ray-tube continuum by Bragg diffraction on a cylindrically bent germanium and silicon crystal in a VOXES-derived von Hamos geometry. The rotation will be coordinated with the axis motors so that the source–crystal–sample geometry and the position of the focused spot are maintained throughout the scan. At each energy set point, a complete fluorescence spectrum will be acquired for a defined dwell time, and $I_f(E)$ will be obtained by fitting or integrating the selected fluorescence line after local-background subtraction. Finer energy steps will be used around the edge and coarser steps in the extended region, while the dwell time will be adapted to the predicted count rate and target statistical uncertainty.

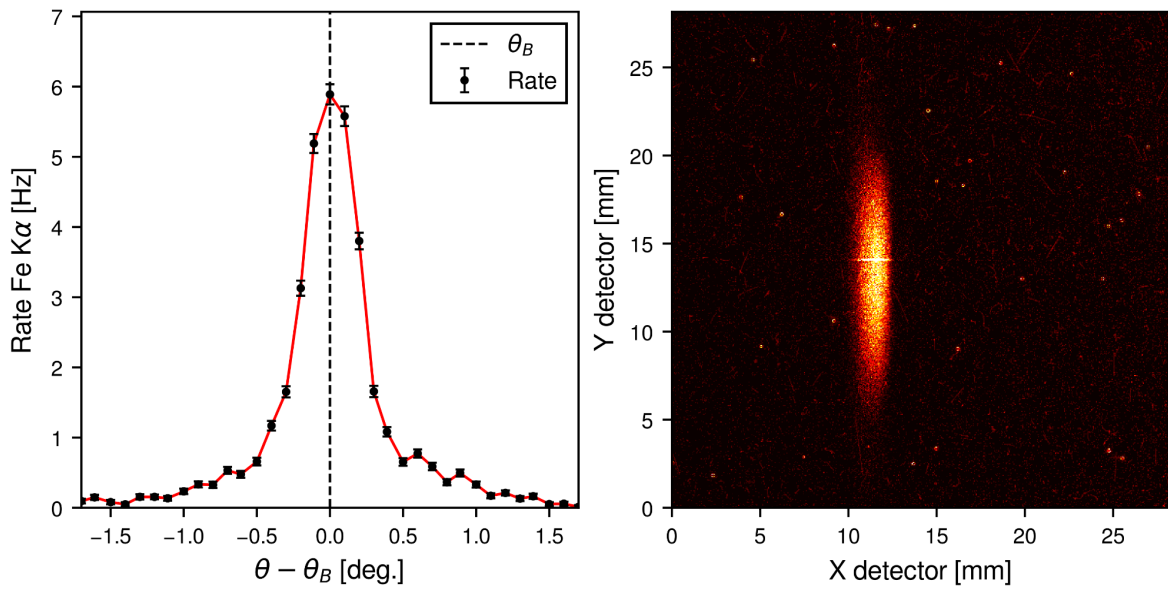


FIGURE 2. Performance of the VOXES setup in angular scanning and X-ray focusing. Left: angular dependence of the Fe $K\alpha$ count rate measured using a mosaic graphite (002) crystal, showing a reflection profile extending over approximately 1° . Right: millimetre-scale Fe $K\alpha$ focal spot recorded with a Timepix detector featuring a $28 \times 28 \text{ mm}^2$ active area.

The feasibility of the required angular scan and focusing is supported by the existing VOXES experience. For instance, Figure 2 (left) shows the measured Fe $K\alpha$ rate on the position-sensitive detector as the crystal is rotated through its nominal Bragg angle, demonstrating a controlled Bragg-angle scan. In Figure 2 (right), the same Fe $K\alpha$ line is shown on a two-dimensional detector, exhibiting millimetric size. Although these measurements were performed in an X-ray-emission configuration and do not themselves constitute XAS, they validate two essential instrumental functions for PRISM: precise crystal rotation for energy selection and spatial focusing with a bent crystal. The project will build on this expertise, the available VOXES components and in-kind contributions, and the shielded laboratory infrastructure at LNF.

To overcome the declining detection efficiency of silicon detectors above 20 keV and the cryogenic requirements of germanium, PRISM will develop a custom multichannel CZT detector system. The design will build on the experience gained within SIDDHARTA, where CZT detectors were first tested in a collider environment, and will be specifically optimized for laboratory fluorescence-XAS.

Unlike commercial CZT detectors, the proposed architecture will be tailored to maximize the collection efficiency of the fluorescence emitted by the sample. A multichannel configuration will provide a larger active area and geometrical acceptance, while segmentation will distribute the counting rate across multiple channels, reducing channel-level pile-up and enabling localized background monitoring and rejection. The baseline design will employ 2-mm-thick CZT crystals with a channel pitch of 500 μm . The detector will be characterized in terms of absolute and relative efficiency, energy resolution, gain linearity and stability, dead time, and maximum usable count rate. The detector will be coupled to dedicated readout electronics capable of recording digitized waveforms for each channel. Waveform-based reconstruction will provide baseline correction, pulse-amplitude estimation, timing extraction, and channel equalization. Pulse-shape-analysis methods employing machine-learning methods, adapted from the experience gained with germanium detectors and currently being applied to CZT tests at BTF (Figure 5), will be used to identify and reject pile-up, saturation, electronic disturbances, and distorted pulses. The same methods will also help mitigate detector-specific effects such as incomplete charge collection, charge trapping, and charge sharing between adjacent channels. Energy calibration, gain matching, efficiency correction, and live-time correction will be performed independently for each channel before the individual spectra are combined into the final fluorescence spectrum.

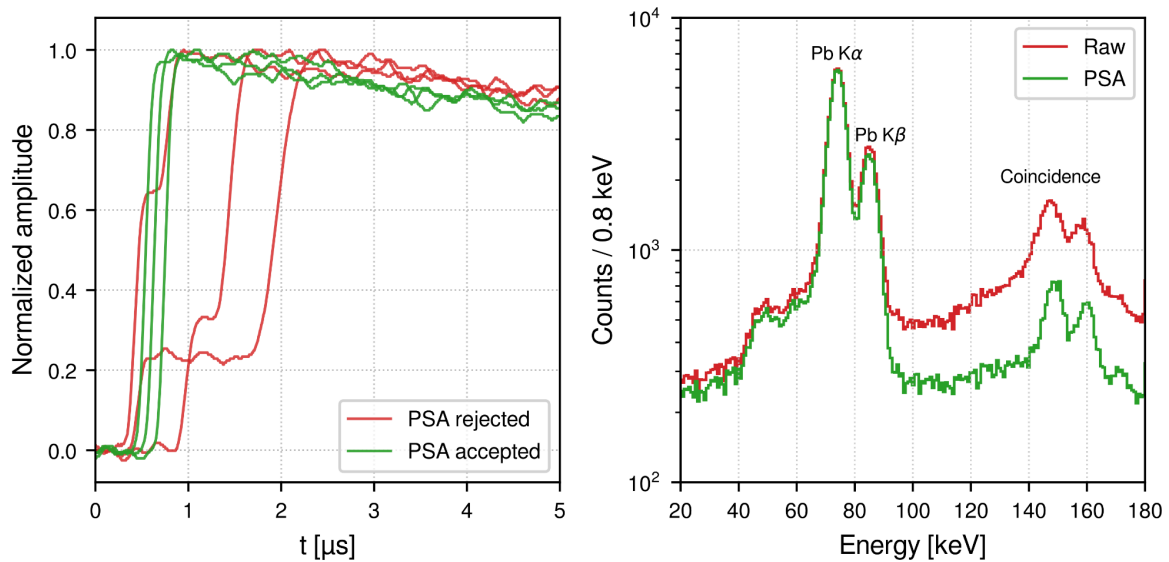


FIGURE 3. Machine-learning-assisted pulse-shape analysis (PSA) of CZT detector signals. Left: representative normalized waveforms accepted by the PSA as clean single-step events (green) and rejected multi-step events (red). Right: calibrated energy spectrum before (Raw) and after PSA selection, showing the preservation of the Pb K α and K β fluorescence peaks and the suppression of the continuum and coincidence structures around 150–160 keV.

The incident intensity $I_0(E)$ will be measured with a motorized exchange system for the sample and CZT head, based on a long-travel linear axis and independent angular positioning. In the fluorescence configuration, the sample will be located at the monochromatic focus and the CZT array will observe it at approximately 90°. In the I_0 configuration, the sample will be translated out of the beam and the CZT detector will be moved and reoriented into the direct monochromatic beam at the focal position; the incident peak recorded by the CZT will then provide $I_0(E)$. Because the direct-beam rate will be

much higher than the fluorescence rate, calibrated attenuators, shorter dwell times and, where necessary, reduced tube current or slit aperture will be used to prevent saturation and pile-up.

To determine the parameters of the components of the setup, within the allocated budget cost, an end-to-end, source-to-detector simulation of the spectrometer will be developed. The Ag K-edge at approximately 25.5 keV will be used as the reference case for defining the XANES measurement protocol and the required performance in terms of energy resolution, photon flux, signal-to-background ratio and acquisition time. The simulation workflow will consist of two coupled stages. First, SHADOW4 ray-tracing simulations building on the approach previously validated for the VOXES spectrometer, which will model photon transport from the X-ray tube through the collimating slits and crystal optics to the sample. Different silicon- and germanium-based crystal reflections will be investigated in the VOXES-type von Hamos geometry. Parametric studies will optimize the crystal material, reflection, curvature radius ρ_c , dimensions, source–crystal and crystal–sample distances, slit apertures and Bragg geometry. The simulations will quantify the geometrical acceptance, energy bandwidth, photon flux, beam profile and illuminated spot size at the sample, while also evaluating alignment and mechanical tolerances. The SHADOW4 output will then be transferred to GEANT4, which will simulate photon absorption in the sample, fluorescence emission, self-absorption, elastic and inelastic scattering, background contributions and energy deposition in the multichannel CZT detector. The combined simulations will be used in the first months of the project to optimize the source, crystal optics, slit system, sample geometry and detector configuration, balancing count rate, spectral resolution and background rejection. A coordinated motorized scanning scheme will also be defined to vary the incident energy across the Ag K-edge while maintaining the required source–crystal–sample geometry. The final outcome of this activity will be a design freeze specifying the X-ray source, crystal type and dimensions, curvature radius, distances, slit apertures, sample position, CZT geometry, motion stages and required positioning accuracy.

A proposed layout of the setup for PRISM is shown in Figure 5, highlighting all the components from the source, the slit system, crystal, sample and the custom CZT detector.

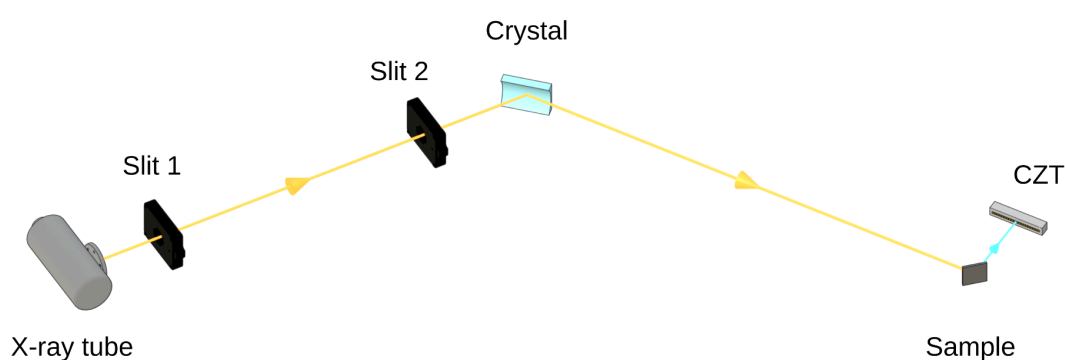


FIGURE 4. Conceptual layout of the proposed PRISM laboratory fluorescence-XAS setup. Radiation from the X-ray tube is collimated by two adjustable slits and monochromatized by a sagittally curved Ge(880) crystal. The diffracted beam illuminates the Ag sample, while the emitted fluorescence is collected off-axis by a linear multichannel CZT detector. The drawing is not to scale, and slit and detector positions are illustrative.

4 Organisation (4000 caratteri)

The PRISM project will have the following organization:

Name	Unit	FTE	Responsibility/WP
Simone Manti	INFN, Section of Frascati	1.0	Scientific Coordinator; overall coordination; WP1–WP4
Alessandro Scordo	INFN, Section of Frascati	0.3	Crystal optics, ray tracing, integration; WP1/WP3
Massimiliano Bazzi	INFN, Section of Frascati	0.3	Electronics/DAQ; WP2
Catalina Curceanu	INFN, Section of Frascati	0.1	Scientific review and application validation; WP4
Totale		1.7	-

It will be divided into the following **Work Packages (WP)**:

- **WP1 – Requirements, simulations and design freeze (M1–M6):** The experimental requirements will be defined using an end-to-end SHADOW4–GEANT4 simulation workflow covering the X-ray source, crystal optics, sample and CZT detector. The resulting specifications will establish the final instrument layout and procurement requirements. Alessandro Scordo will contribute his expertise in crystal optics, ray tracing and VOXES-based spectrometer design.
- **WP2 – CZT, front-end, DAQ, calibration and characterization (M4–M15):** A multichannel CZT detector and its dedicated low-noise front-end electronics and waveform-recording DAQ will be designed and developed. Channel-by-channel calibration, gain equalization, efficiency measurements and characterization of energy resolution, stability, dead time and pile-up will be performed. Pulse-reconstruction and pulse-shape-analysis methods will be implemented to improve spectral quality and reject distorted events. Massimiliano Bazzi will support the development and validation of the electronics and DAQ chain.
- **WP3 – Procurement, mechanics, integration and commissioning (M10–M18):** Following the design freeze, the X-ray source, crystal optics, motion stages, slits and mechanical components will be procured and subjected to acceptance tests. The source, monochromator, sample positioning system and CZT detector will then be mechanically and electronically integrated. Commissioning will include alignment, synchronized energy scanning, control-system validation and comparison of the measured flux, spot size and energy resolution with the simulation predictions.
- **WP4 – Ag reference samples and realistic sample demonstration (M16–M24):** The complete instrument will first be validated through Ag-foil measurements and Ag reference compounds representing different chemical states. Quantitative XANES performance will be assessed in terms of energy calibration, reproducibility, signal-to-noise ratio and agreement with reference spectra. Fluorescence-XAS will subsequently be demonstrated on a realistic thick, diluted or heterogeneous sample, and the accessible operating domain will be mapped.

Alessandro Scordo will support the measurements, crystal-optics optimization and interpretation of the experimental results.

The project **Milestones (MS)** are:

- **MS1 – Instrument design frozen (M6).** Experimental requirements, complete source–optics–sample–detector geometry and procurement specifications approved.
- **MS2 – CZT detection chain validated (M15).** Intrinsic efficiency above 90% at 40 keV, energy resolution of 1 keV FWHM or better, dead time below 10% and residual pile-up below 5% at the selected operating rate.
- **MS3 – Integrated instrument commissioned (M18).** Synchronized scanning and energy calibration operational; measured rate within 20%, and spot size and energy resolution within 10% of simulation predictions.
- **MS4 – Fluorescence XAS capability demonstrated (M24).** Ag K-edge XANES obtained with $\text{SNR} \geq 20$ and edge-position reproducibility within 1 eV; two chemical states discriminated in a realistic sample with a significance of at least 3σ ; operating domain documented.

The Work Packages and the Milestones are summed in the Gantt table in Figure 6:

WORK PACKAGE (WP)	PLANNED ACTIVITY (A)	2027				2028			
		1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24
WP1 - Requirements, simulations and design freeze	A1.1 - Requirements definition	■							
	A1.2 - SHADOW4 ray-tracing optimization	■							
	A1.3 - GEANT4 fluorescence simulations		■						
	A1.4 - Final geometry selection		■	◆ MS1					
WP2 - CZT, front-end, DAQ, calibration and characterization	A2.1 - CZT detector and front-end design		■	■					
	A2.2 - Multichannel DAQ and signal-processing development			■	■				
	A2.3 - CZT calibration and performance characterization				■	■			
	A2.4 - Detector validation and operating-point selection					■	◆ MS2		
WP3 - Procurement, mechanics, integration and commissioning	A3.1 - Procurement of source, optics and motion stages				■	■			
	A3.2 - Mechanical design and spectrometer assembly				■	■			
	A3.3 - Hardware, software and control-system integration				■	■			
	A3.4 - Instrument commissioning and verification					■	◆ MS3		
WP4 - Ag reference samples and realistic sample demonstration	A4.1 - Ag foil calibration and reproducibility tests					■	■		
	A4.2 - XANES validation on Ag reference compounds					■	■	■	
	A4.3 - Fluorescence XAS demonstration on a realistic sample						■	■	■
	A4.4 - Performance map and operating-domain assessment							■	◆ MS4

FIGURE 5. Gantt table of the PRISM objected divided in each Work Packages (WPs) and for each one its Planned Activity (A) through the two years and Milestones (MS) are indicated by the yellow mark.

The project **Deliverables (D)** are:

- **D1 – Validated SHADOW4–GEANT4 workflow, frozen instrument design and procurement specifications (M6).**
- **D2 – CZT calibration and performance-characterization dataset with acceptance report (M15).**
- **D3 – Ag foil and reference-compound XANES dataset with quantitative validation report (M21).**

- **D4** – Realistic-sample dataset, operating-domain assessment, documented analysis tools and final project report (M24).

5 Synergies with Other Activities (3000 caratteri)

The PRISM project will establish extensive synergies with various external projects, institutions and companies.

Other INFN Scientific Committee

- **CSN3:** PRISM has direct synergies with KAONNIS, building on the experience gained with CZT detectors in the SIDDHARTA experiment at DAΦNE. Additional synergies are expected with the VIP experiment at LNGS, which faces similar challenges in the detection of weak X-ray fluorescence lines above 20 keV.
- **CSN5:** The strongest connection is with VOXES, a former CSN5 Young Researchers project whose expertise in crystal-based X-ray spectroscopy, instrumentation and data analysis provides the technological foundation for PRISM. VOXES will contribute existing equipment and laboratory infrastructure in kind, including facilities for testing and calibration with laboratory X-ray sources. Further synergies are foreseen with SPHINX, particularly in precision angular scanning, optical alignment and the integration of X-ray optics and detectors.
- **EuPRAXIA:** Future synergies are expected with EuAPS and the AQUA beamline, both targeting XAS measurements in different photon-energy ranges. Developing laboratory-based XAS methods within PRISM will provide complementary expertise, instrumentation and procedures useful for preparing and optimizing future experiments at these facilities.

External institutions and national and international research laboratories

- **University of Palermo:** PRISM will benefit from the established collaboration with the University of Palermo in the development and testing of CZT detectors within the SIDDHARTA experiment. This synergy will support the optimization of the detector geometry, front-end electronics, calibration procedures and signal-reconstruction methods.
- **ENRICH COST Action:** PRISM is strongly aligned with the ENRICH COST Action, which promotes a European network for radiation-detection research and its applications to societal challenges. Within this network, the applicant is co-leader of the Working Group on Simulation. A particularly relevant connection is with Manuel Sánchez del Río, leader of the Working Group and principal developer of SHADOW4, the ray-tracing software that will be used to optimize the crystal optics and spectrometer geometry.

Industry and Private Sector

The following companies have provided letters of support for PRISM, confirming industrial interest in the proposed instrumentation and its technology-transfer potential.

- **Beamide:** Beamide may support the validation and industrial exploitation of the GEANT4-based radiation-transport framework.

- **Due2Lab:** Due2Lab will contribute expertise in CZT crystal production and detector integration, strengthening the feasibility and scalability of the multichannel system.
- **Rigaku:** Rigaku will provide an industrial benchmark for laboratory XAS and support the selection of X-ray sources, optics, and key components.

6 Impact and Spin-offs (3000 caratteri)

PRISM will establish at INFN-LNF a validated laboratory capability for quantitative fluorescence-XAS above 20 keV based on room-temperature multichannel detection. By combining crystal optics, CZT sensors, low-noise electronics and DAQ with SHADOW4-GEANT4 modelling, PRISM will address a regime where efficient non-cryogenic laboratory spectroscopy remains challenging.

For the INFN community, the impact will extend beyond a single instrument. PRISM will valorize the existing VOXES infrastructure and consolidate LNF expertise in semiconductor radiation detectors, precision X-ray spectroscopy, radiation transport and spectral reconstruction. Its modular CZT detection and readout chain, validated source-to-detector models, calibration protocols and performance maps will be reusable by other INFN activities developing hard-X-ray instrumentation. The project will establish a transferable route from predictive simulation and subsystem tests to quantitative validation of an integrated instrument. After laboratory validation, its design rules may inform future facilities such as EuAPS. PRISM will also train young researchers and technical staff across optics, detectors, electronics, simulation and data analysis.

For the wider scientific community, PRISM will expand laboratory access to fluorescence-XAS at the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd. If its quantitative targets are met, the platform will support sample screening, method development, cell optimization and long in-situ or operando studies of thick, dilute and heterogeneous specimens that are difficult to investigate in transmission. PRISM will complement, not replace, synchrotrons and XFELs: preliminary measurements and protocol optimization can make scarce beamtime more effective, while repeated scans and slowly evolving processes can remain in the laboratory. Beneficiaries include catalysis, electrochemistry, energy materials and advanced alloys. Documented workflows, reference datasets and operating-domain maps will improve reproducibility and help other laboratories assess the approach.

The main technology-transfer opportunities are the modular room-temperature CZT detector, front-end and multichannel DAQ chain; its calibration and pulse-reconstruction methods; the validated source-optics-sample-detector workflow; and the compact hard-X-ray spectrometer architecture. Following experimental validation, these assets may be assessed with INFN technology-transfer structures and X-ray instrumentation companies for protection or co-development. Potential applications include routine materials characterization, quality control and long-duration monitoring of catalysts, batteries and advanced materials. Industrial uptake will remain conditional on demonstrated performance, robustness, manufacturability, cost and market need.

7 Costs

The overall cost of PRISM is 112 k€, where 66 k€ are requested for 2027 and 46 k€ for 2028. The in-kind contribution from VOXES

Year	Category	Item	Purpose/WP	Cost [€]
2027	Equipment	CZT Detector and Front-End	Fluorescence Detection - WP2	€30.000
2027	Equipment	Digitizer and DAQ	Signal Acquisition - WP2	€20.000
2027	Equipment	Motorized Stages and Controllers	Automated Positioning - WP3	€10.000
2027	Consumables	Ag Reference Targets	Calibration Tests - WP2/WP4	€2.000
2027	Services	Custom Mechanical Supports	Instrument Integration - WP3	€2.000
2027	Computing	DataCloud CPU and Storage	End-to-End Simulations - WP1	€2.000
Total 2027				€66.000
2028	Equipment	Cooled X-ray Tube	Excitation Source - WP3	€10.000
2028	Equipment	X-ray Tube Power Supply	Source Operation - WP3	€10.000
2028	Travel	Oral Conference Travel	Results Dissemination - WP4	€4.000
2028	Equipment	Bent Crystal Optics	Energy Selection - WP3	€20.000
2028	Consumables	Additional Reference Targets	Multi-Edge Validation - WP4	€2.000
Total 2028				€46.000
Total				€112.000

TABLE I. Detailed financial request for the two-year PRISM project, broken down by year, cost category, item, and related Work Package.

For the **equipment**:

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8 Risk Assessment

Here I will outline the possible risks (R) in the project in relation with the specific planned activity and describe the possible mitigation.

R1. Delayed delivery of critical components (T2.2–T2.4)

- **Risk:** Delays in the delivery of crystal optics, mechanics or detector components may postpone integration and commissioning.
- **Mitigation:** Use compatible in-kind VOXES components, including non-optimized crystals, for preliminary integration and commissioning until the final components become available.

R2. Slow CZT Optimization (T3.1–T3.3)

- **Risk:** Calibration and optimization of the CZT detector and electronics may require longer than planned, delaying laboratory XAS validation.
- **Mitigation:** Use an in-kind VOXES Si PIN diode to start calibration and reference scans, with reduced efficiency, while CZT optimization continues in parallel.

R3. Inaccuracy of End-to-End Simulations (T1.2–T1.4)

- **Risk:** SHADOW4–GEANT4 simulations may not accurately reproduce the source, crystal, sample and detector response, affecting the final geometry selection.
- **Mitigation:** Benchmark the model against analytical calculations and VOXES measurements and maintain direct contact with SHADOW4 developer Manuel Sánchez del Río through COST Action ENRICH WG3.

R4. Excessive XANES acquisition time (T3.3, T4.2–T4.3)

- **Risk:** Insufficient photon flux or signal-to-background ratio may make XANES measurements longer than planned.
- **Mitigation:** Optimize scan steps and dwell times and, if necessary, widen the optical slits and shorten the CZT peaking time within the minimum resolution and background-rejection requirements.

9 References

- [1] Zaera, F. “*In-situ* and *operando* spectroscopies for the characterization of catalysts and of mechanisms of catalytic reactions.” *J. Catal.*, **404**, 900–910, 2021. doi: 10.1016/j.jcat.2021.08.013.
- [2] Bonn, M., Domke, K. F., Sulpizi, M., McCrory, C., and Peng, Z. “In situ and operando characterization: Advancing our understanding of functional materials.” *J. Chem. Phys.*, **162**(14), 140402, 2025. doi: 10.1063/5.0270688.
- [3] Yang, Y., Feijóo, J., Briega-Martos, V., Li, Q., Krumov, M., et al. “*Operando* methods: A new era of electrochemistry.” *Curr. Opin. Electrochem.*, **42**, 101403, 2023. doi: 10.1016/j.coelec.2023.101403.
- [4] Fehse, M., Iadecola, A., Simonelli, L., Longo, A., and Stievano, L. “The rise of X-ray spectroscopies for unveiling the functional mechanisms in batteries.” *Phys. Chem. Chem. Phys.*, **23**(41), 23445–23465, 2021. doi: 10.1039/D1CP03263A.
- [5] Bunker, G. *Introduction to XAFS: A Practical Guide to X-ray Absorption Fine Structure Spectroscopy*. Cambridge: Cambridge University Press, 2010. doi: 10.1017/CBO9780511809194.
- [6] Calvin, S. *XAFS for Everyone*. Boca Raton: CRC Press, 2013. doi: 10.1201/b14843.
- [7] Bès, R., Ahopelto, T., Honkanen, A.-P., Huotari, S., Leinders, G., et al. “Laboratory-scale X-ray absorption spectroscopy approach for actinide research: Experiment at the uranium L3-edge.” *J. Nucl. Mater.*, **507**, 50–53, 2018. doi: 10.1016/j.jnucmat.2018.04.034.
- [8] Zimmermann, P., Peredkov, S., Abdala, P. M., DeBeer, S., Tromp, M., et al. “Modern X-ray spectroscopy: XAS and XES in the laboratory.” *Coord. Chem. Rev.*, **423**, 213466, 2020. doi: 10.1016/j.ccr.2020.213466.
- [9] Scordo, A., Curceanu, C., Miliucci, M., Shi, H., Sirghi, F., et al. “VOXES: a high precision X-ray spectrometer for diffused sources with HAPG crystals in the 2–20 keV range.” *J. Instrum.*, **13**(04), C04002, 2018. doi: 10.1088/1748-0221/13/04/C04002.
- [10] Manti, S. and Scordo, A. “Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies.” *Eng. Innov.*, **19**, 23–29, 2026. doi: 10.4028/p-rQpe9Z.
- [11] Yamamoto, T. “Selection of Monochromator Crystal for Pb L-, Br K-, and Ag K-Edge XANES Using a Laboratory-Based Spectrometer.” *X-Ray Spectrom.*, **55**(4), 538–544, 2026. doi: 10.1002/xrs.70076.
- [12] Del Rio, M. S., Reyes-Herrera, J., Shi, X., and Rebuffi, L. “SHADOW4: the popular ray tracing revived for evolving synchrotron sources in fourth-generation storage rings.” *J. Phys. Conf. Ser.*, **3010**(1), 012071, 2025. doi: 10.1088/1742-6596/3010/1/012071.
- [13] Scordo, A., De Leo, V., Curceanu, C., Miliucci, M., and Sirghi, F. “Efficiency measurements and simulations of a HAPG based Von Hamos spectrometer for large sources.” *J. Anal. At. Spectrom.*, **36**(11), 2485–2491, 2021. doi: 10.1039/d1ja00214g.